

# DATASHEET

## Variable Speed Drives



Product coding : CFW500B03P0T5DB66DS  
 Product reference : 14978915  
 Accessory module (control) : CFW500

Power supply : 500-600 V  
 Input minimum-maximum voltage :  
 - In :  
 - Out :

|                             |                      |                     |
|-----------------------------|----------------------|---------------------|
| Supply voltage range        | 500-600 V            |                     |
| Overload cycle              | Normal Overload (ND) | Heavy Overload (HD) |
| Rated current (HD)          | 3                    | 3                   |
| CORRENTE_SOBRECARGA_60S_CFW |                      |                     |
| CORRENTE_SOBRECARGA_3S_CFW  |                      |                     |

Maximum applicable motor:

| Voltage/Frequency | Power (HP/kW) [1]    |                     |
|-------------------|----------------------|---------------------|
|                   | Normal Overload (ND) | Heavy Overload (HD) |

Accessory module (control) :  
 Dynamic braking [2] : Standard with braking  
 External electronic supply 24Vcc : Not available  
 Safety Stop : Not available  
 Internal RFI filter : Without filter  
 :  
 Link Inductor :  
 Memory card : Not included in the product  
 USB port : Only with plug-in  
 Line frequency : 50/60Hz  
 Line frequency range (minimum - maximum) : 48-62 Hz  
 Phase unbalance : Less or equal to 3% of input rated line voltage  
 Transient voltage and overvoltage : Category III  
 Single-phase input current [3] : Not applicable  
 Three-phase input current [3] : 3,7 A  
 Power factor :  
 Displacement factor : 0,98  
 Rated efficiency :  
 Maximum connections (power up cycles - on/off) per hour : 10 (1 each 6 minutes)  
 DC power supply :  
 Standard switching frequency :  
 Selectable switching frequency : 2,5 and 15 kHz  
 Real-time clock : Not available  
 COPY Function : Yes, by MMF  
 Dissipated power:

| Mounting type | Overload       |                |
|---------------|----------------|----------------|
|               | ND             | HD             |
| Surface       |                |                |
| Flange        | Not applicable | Not applicable |

### Source available to the user

Output voltage :  
 Maximum capacity :  
 Power supply : Switched-mode power supply  
 Control method : V/f, VVW, Sensorless and Encoder  
 Encoder interface :  
 Control output frequency : 0-500 Hz  
 Frequency resolution : 0,015 Hz  
 - Speed resolution : 1% of rated speed  
 - Speed range : 1:20  
 - Speed resolution : 1% of rated speed  
 - Speed range : 1:30  
 - Speed resolution : 0,5% of rated speed  
 - Speed range : 1:100  
 - Speed resolution : 0,1% of nominal speed  
 - Speed range : Up to 0 rpm  
 Quantity (standard) :

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|                                          |                  |
|------------------------------------------|------------------|
| Levels                                   | :                |
| Impedance for voltage input              | :                |
| Impedance for current input              | :                |
| Function                                 | :                |
| Maximum allowed voltage                  | :                |
| Quantity (standard)                      | :                |
| Activation                               | :                |
| Maximum low level                        | :                |
| Minimum high level                       | :                |
| Input current                            | :                |
| Maximum input current                    | :                |
| Function                                 | :                |
| Maximum allowed voltage                  | :                |
| <b>Analog outputs</b>                    |                  |
| Analogic outputs - Quantity (standard)   | :                |
| Levels                                   | :                |
| RL for voltage output                    | :                |
| RL for current output                    | :                |
| Function                                 | :                |
| <b>Digital outputs</b>                   |                  |
| Digital outputs - Quantity (standard)    | :                |
| Maximum voltage                          | :                |
| Maximum current                          | :                |
| Function                                 | :                |
| -                                        |                  |
| -                                        |                  |
| -                                        |                  |
| -                                        |                  |
| -                                        |                  |
| -                                        |                  |
| -                                        |                  |
| -                                        |                  |
| -                                        |                  |
| -                                        |                  |
| <b>Available protection</b>              |                  |
| - Output phase-phase overcurrente/Short  |                  |
| - Overcurrent/Short circuit phase-ground |                  |
| - Under/Overtoltage in power             |                  |
| - Heat sink overtemperature              |                  |
| - Motor overload                         |                  |
| - IGBT's modules overload                |                  |
| - Fault/External alarm                   |                  |
| - Programming error                      |                  |
| <b>Operation interface (HMI)</b>         |                  |
| Avaliability                             | :                |
| Installation                             | :                |
| Number of HMI buttons                    | :                |
| Display                                  | :                |
| Indication accuracy                      | :                |
| Speed resolution                         | :                |
| Standard HMI degree of protection        | :                |
| HMI battery type                         | : Not applicable |
| HMI battery life expectancy              | : Not applicable |
| Remote HMI type                          | : Accessory      |
| Remote HMI frame                         | : Not applicable |
| Remote HMI degree of protection          | : IP54           |
| Enclosure                                | : IP66           |
| Degree of pollution                      | : 2              |
| RoHS                                     | : Yes            |
| Conformal Coating                        | :                |
| - Size                                   | : 00009          |
| - Height                                 | :                |
| - Width                                  | :                |
| - Depth                                  | :                |
| - Weight                                 | :                |
| <b>Mechanical Installation</b>           |                  |
| Mounting position                        | :                |

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### Mechanical Installation

Fixing screw :  
 Tightening torque :  
 Allows side-by-side assembly :  
 Minimum spacing around the inverter:  
 - Top :  
 - Bottom :  
 - Front :  
 - Side :

Cable gauges and tightening torques:

|           | Recommended cable gauge | Recommended tightening torque |
|-----------|-------------------------|-------------------------------|
| Power     |                         |                               |
| Braking   |                         |                               |
| Grounding |                         |                               |
| Control   |                         |                               |

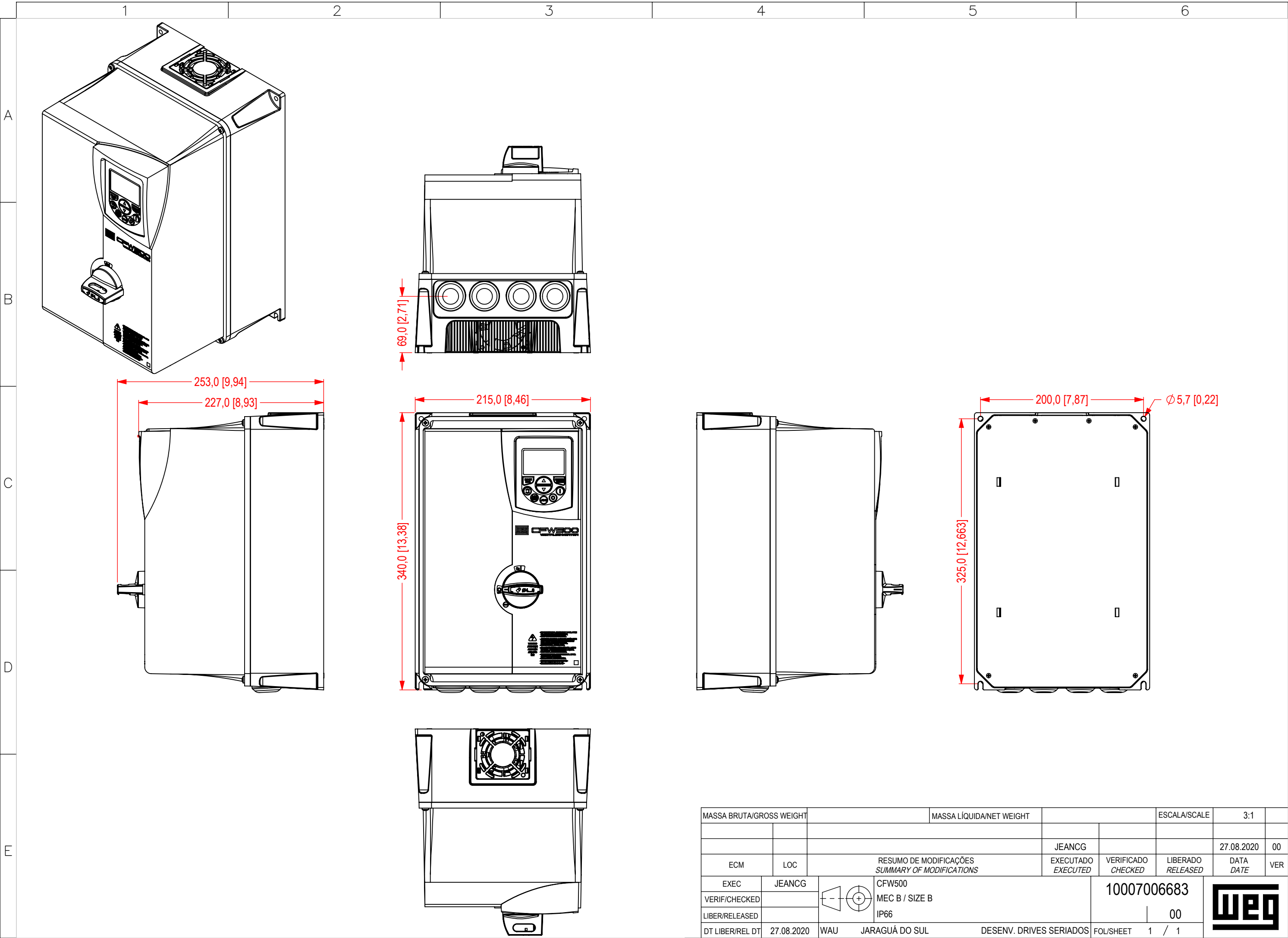
SoftPLC : Yes, incorporated  
 Maximum breaking current :  
 Minimum resistance for the brake resistor :  
 Recommended aR fuse :  
 Recommended circuit breaker :  
 Disconnect switch : With disconnect switch  
 Motor coupling box : Not applicable

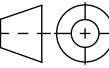
### Standards

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Safety                        | <ul style="list-style-type: none"> <li>- UL 508C - Power conversion equipment.</li> <li>- UL 840 - Insulation coordination including clearances and creepage distances for electrical equipment.</li> <li>- EN 61800-5-1 - Safety requirements electrical, thermal and energy.</li> <li>- EN 50178 - Electronic equipment for use in power installations.</li> <li>- EN 60204-1-Safety of machinery. Electrical equipment of machines. Part 1: General requirements. Note: To have a machine in accordance with that standard, the manufacturer of the machine is responsible for the installation of an emergency-stop device and a network switching equipment.</li> <li>- EN 60146 (IEC 146) - Semiconductor converters.</li> <li>- EN 61800-2 - Adjustable speed electrical power drive systems - Part 2: General requirements - Rating specifications for low voltage adjustable frequency AC power drive systems.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Electromagnetic Compatibility | <ul style="list-style-type: none"> <li>- EN 61800-3 - Adjustable speed electrical power drive systems - Part 3: EMC product standard including specific test methods.</li> <li>- EN 55011 - Limits and methods of measurement of radio disturbance characteristics of industrial, scientific and medical (ISM) radio-frequency equipment.</li> <li>- CISPR 11 - Industrial, scientific and medical (ISM) radio-frequency equipment</li> <li>- Electromagnetic disturbance characteristics - Limits and methods of measurement.</li> <li>- EN 61000-4-2 - Electromagnetic compatibility (EMC) - Part 4: Testing and measurement techniques - Section 2: Electrostatic discharge immunity test.</li> <li>- EN 61000-4-3 - Electromagnetic compatibility (EMC) - Part 4: Testing and measurement techniques - Section 3: Radiated, radio-frequency, electromagnetic field immunity test.</li> <li>- EN 61000-4-4 - Electromagnetic compatibility (EMC) - Part 4: Testing and measurement techniques - Section 4: Electrical fast transient/burst immunity test.</li> <li>- EN 61000-4-5 - Electromagnetic compatibility (EMC) - Part 4: Testing and measurement techniques - Section 5: Surge immunity test.</li> <li>- EN 61000-4-6 - Electromagnetic compatibility (EMC)- Part 4: Testing and measurement techniques - Section 6: Immunity to conducted disturbances, induced by radio-frequency fields.</li> </ul> |
| Mechanical Construction       | - EN 60529 e UL 50                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

### Notes

- 1) Motor power is orientative, valid for standard WEG Motors of IV poles. The correct sizing must be done according to the nominal current of the motor used, which must be less than or equal to the rated output current of the inverter;
- 2) Braking resistor is not included;
- 3) Considering minimum line impedance of 1%;
- 4) For more information, refer to the user manual of CFW500;
- 5) All images are merely illustrative.
- 6) For operation with switching frequency above nominal, apply derating to the output current (refer to the user manual).



|                          |            |                                                                                       |                |                         |                       |                      |
|--------------------------|------------|---------------------------------------------------------------------------------------|----------------|-------------------------|-----------------------|----------------------|
| MASSA BRUTA/GROSS WEIGHT |            | MASSA LÍQUIDA/NET WEIGHT                                                              |                | ESCALA/SCALE            | 3:1                   |                      |
|                          |            |                                                                                       |                |                         |                       |                      |
|                          |            |                                                                                       | JEANCG         |                         | 27.08.2020            | 00                   |
| ECM                      | LOC        | RESUMO DE MODIFICAÇÕES<br>SUMMARY OF MODIFICATIONS                                    |                | EXECUTADO<br>EXECUTED   | VERIFICADO<br>CHECKED | LIBERADO<br>RELEASED |
| EXEC                     | JEANCG     |  |                | 10007006683             |                       |                      |
| VERIF/CHECKED            |            |                                                                                       |                | 00                      |                       |                      |
| LIBER/RELEASED           |            |                                                                                       |                |                         |                       |                      |
| DT LIBER/REL DT          | 27.08.2020 | WAU                                                                                   | JARAGUÁ DO SUL | DESENV. DRIVES SERIADOS | FOL/SHEET 1 / 1       |                      |